

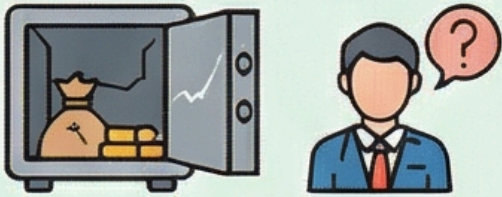
High Value Asset / Inventory Risk Oracle

A Hardware-Anchored Assessment System



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Introduction: Ensuring Collateral Integrity through Real-Time Visibility



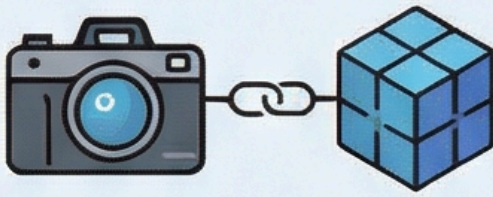
Vulnerability of High-Value Reserves

Traditional management relies on intermittent human audits, creating a gap where fraud, such as the **Qingdao Metals Scandal** can occur unnoticed until the collateral has already been moved or pledged multiple times.



Deterministic Smart Contract Logic

By replacing human oversight with smart contracts, the system ensures that predefined actions like alerts or liquidations occur automatically based on verified sensor data.



Real-Time Visual Proof

Integrating visual evidence (photos) with weight data provides an immutable **"proof of reserve"** on the blockchain that can be audited by lender party at any time.

Methods: The 9-Step Risk Oracle Cycle



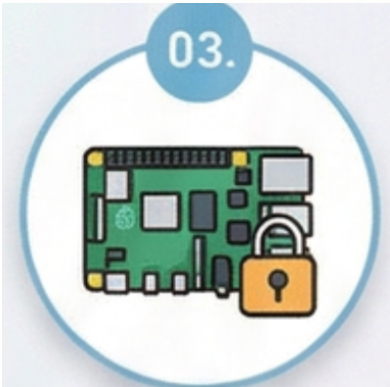
Baseline Establishment

Initial inventory weight is measured and set as the "Pledged Baseline" within the smart contract



Continuous IoT Monitoring

A weight sensor monitors the collateral mass in real-time to detect any tampering.



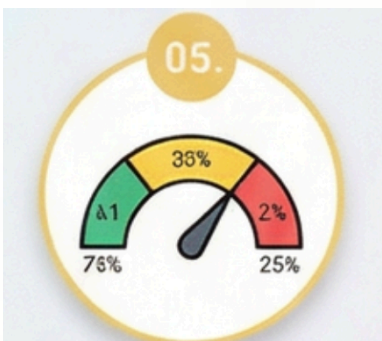
Cryptographic Signing

Acting as a secure gateway, the RaspberryPi authenticates sensor data through digital signing to mitigate manipulation risks



Oracle Transmission

The Oracle script pushes the signed hardware data to the Sepolia Testnet blockchain.



Multi-Stage Threshold Detection

The system monitors collateral weight against 3 checkpoints: 75% (L1), 50% (L2), and 25% (L3) of the expected reserve.



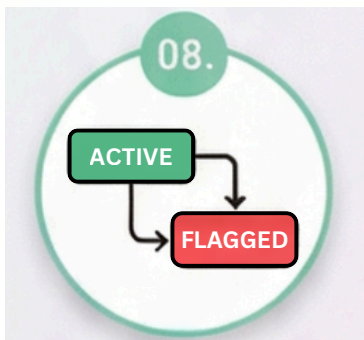
Sensor-Triggered Visual Capture

Upon detection of a weight change, the camera captures a real-time photo for immediate visual verification.



On-Chain Logic Evaluation

The smart contract compares the weight ratio against set loan-to-value (LTV) ratio.



Contract State Transition

The contract moves from **ACTIVE** to **FLAGGED** when the collateral value drops below the risk threshold.



Automated Escalation & Alert

The system emits an alert to the web dashboard and then human interference is needed for L1/ L2/ L3 Escalation levels

Results



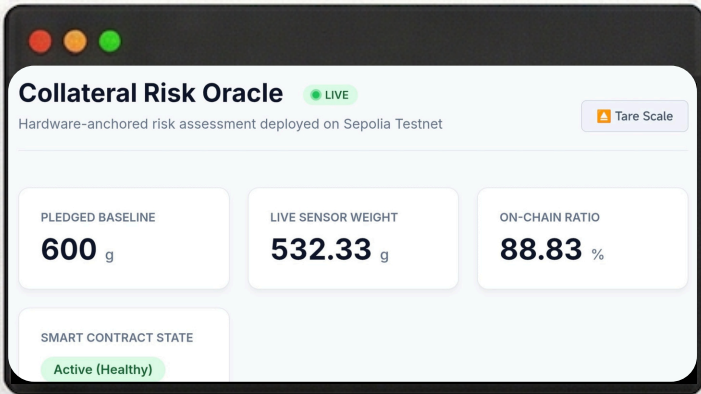
Weight to Digital Integrity

Physical changes are converted into digital triggers ensuring blockchain state reflects the physical reality of the vault.

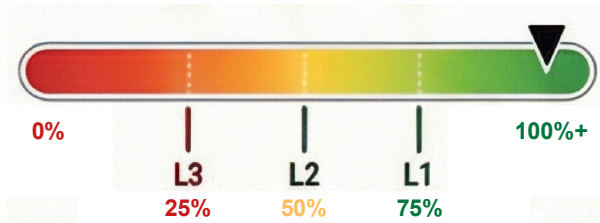
Pseudocode

```
function reportWeight (uint256 newWeight, bytes32 proofHash) external {
    currentWeight = newWeight;
    uint256 ratio = (newWeight * 100) / pledgedWeight;
    if (ratio < 25) {
        currentState = State.FLAGGED_L3;
    } else if (ratio < 50) {
        currentState = State.FLAGGED_L2;
    } else if (ratio < 75) {
        currentState = State.FLAGGED_L1;
    } else {
        currentState = State.ACTIVE;
    }
}
```

Dashboard



Collateral Health



Latest Anchored Event

```
{
  "image": "static/capture_1771895956.jpg",
  "timestamp": 1771895962,
  "weight": 533.14
}
```

Proof Hash: da8400ca03334b6a0313345a46201158ed

Discussions & Conclusions



Reducing Collateral Risk

Automated monitoring minimizes the risk of theft or unauthorized pledging that often goes unnoticed in manual systems.



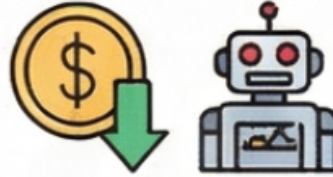
Blueprint for Trustless Custody

This model provides a foundation for high-value asset lending in decentralized finance by removing the need for a trusted third-party



Deterministic Process Discipline

Blockchain ensures that the monitoring protocol is followed exactly as intended, removing human bias or corruption from the audit process.



Reducing Monitoring Costs

Automating the audit trail significantly lowers the overhead associated with manual physical inspections of reserves

References

- World Bank Group (2020). Smart Contract Technology and Financial Inclusion. Fintech Note No. 6. Washington, DC: World Bank.
- Antonopoulos, A. M., & Wood, G. (2018). Mastering Ethereum: Building smart contracts and DApps. O'Reilly Media.